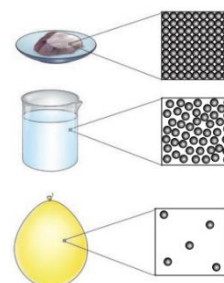
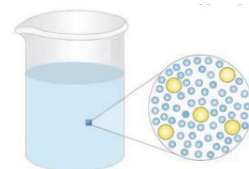


Chapter – 1: Exploring the Investigative World of Science

- Our journey – *curiosity* – science world – continue – grade 8
- Hope – you bring – spirit – adv., exploration
- Grade 6 – discover beginning – science – why, how – world around us
- Grade 7 – learnt – science – evolving daily – each ans. – new ques. – ideas – change – explore deeper
- Grade 8 – next step – investigative world – science – wonder, evolution – come together – form heart – how science works
- We want – learn new facts – BUT also – learn – how to find new facts
- Investigation – science – means more – just looking, asking simple ques.
- Now – ask – more focused ques. – design ways – simple exp. – ans. ques. – use obs. – improve understanding
- Step by step – learn – use ques. – start point – obs., exp., explain – what we see
- Doing so – you – learners – BUT also – investigators, young scientists – explore – real world puzzles
- These puzzles – range – everyday life – why dough rise – bigger mysteries – Earth, beyond – global warming
- Turn each page – this book – notice design – page no.
- Left-hand page – bottom – root – symbolize – deep, solid foundation – knowledge – connected – env., tradition, culture, natural heritage
- Right-hand page – top corner – kite – remind us – curiosity – take flight – explore unknown
- Together – 2 symbols – root, kite – invite you – stay grounded – real obs. – allow ideas – soar – new horizons
- Remember – investigation – science – works best – balance – solid ground – careful obs. – freedom – creative thinking
- Also notice – some patterns – lines – bottom – hidden science
- Brief look – various stops – our journey – see – our curiosity – take us
- This yr. – investigate adv. – journey – tiny microbes – can't see – TO – planet-wide challenges – can't ignore
- Start – examine – small drops of water – uncover – hidden world – tiny org. – unseen – BUT – deep link – us
- Some – invisible helpers – digest food, produce medicines – others – harmful – cause infections
- Our body – need what – stay healthy? – how – fight infections?
- Find out – nutritious food, exercise, medication, vaccine – help – stay healthy, fight infections
- BUT – this – just beginning – today's world – science – major role – improve lives
- Ex. – elec. current – many ways – life easier
- Depend – heating effect – elec. current – keep warm – magnetic effect – motors, machines – func.
- Phenomenon – depend – fundamental forces – after elec. – move on – study forces – make obj. – speed up, slow down, change dir.
- Understand force – help explain – ball – thrown up – air – falls back – OR – car stops – brakes applied
- Also lead us – idea – pressure – force – distributed over obj.
- Same concept – force, pressure – decide – air movement
- Small diff. – pressure – gentle breeze – stronger diff. – strong winds, even cyclones
- These forces – connected – powerful weather events – storms, cyclones – affect – daily lives, agriculture, safety
- Truly understand – air – exert pressure – OR – water boils – certain temp. – zoom in – materials – see – what particles – made of – how – move around



- Everything around – tiny particles – materials – solid – particles – cannot move much – gases – move more freely
- Classifying things – imp. feature – science
- Classify materials – elements, compounds, mixtures
- We know – particles – combine, mix – understand solu. – sugar in tea – makes sweet
- World – particles, mixture – move into – world – light
- Study – light rays – reflect off – flat, curved mirrors – bend – passing through lens – help understand – working – things – around us
- Bending of light – explain – img. – shiny metal spoon – OR – corrective glasses – see clearly
- Polished mirror – reflect light – BUT also – rough surfaces – even Moon
- Depend – rel. pos. – Earth, Moon, Sun – slightly diff. part – Moon – illuminated each night – give rise – beautiful phases – Moon



- Watch – periodic cycle – Moon phases – humans – 1st calendars
- Combine – obs. – sunrises, sunsets, lunar cycles – various calendars
- Fascinating – calendars – determine routines – Earth – link to – motions – obj. – far beyond
- Not only – calendars, movements – Sun, Moon – link
- Here – Earth – marvellous, complex patterns – rel. b/w – living org., their env.
- Every living being – insect to whale – grass to trees – depend, respond – air, water, sunlight, other org. – form ecosys. – support life
- Final chapter – put together – understand – makes Earth – right for life – recognise – urgent challenges – planet faces
- Most imp. – Earth – perfect dist. – Sun – water – liquid – atm. – provide oxygen – shield us – UV rays
- Human act. – planet – cause changes – temp. – Earth – disrupt climate – dangerous consequences
- Heart – both – problem, possible solu. – us
- We – influence climate – BUT – we – also can, must – use science – understand changes – guide actions
- Same scientific principles – guided journey – middle stages – obs., measure, exp. – key – help protect – delicate balance – life depends

Why is one side of a *puri* thinner than other?

- Remember – science – everywhere – no need – fancy lab. – simple exp.
- Kitchen – home – wonderful place – obs., ask ques.
- Start – curiosity – careful obs. – ask – what if ?
- Notice – *puri*, *bhatura* – puffs up – hot oil – OR – *phulka* – swells – flame
- Why – puff up – one side – thinner? – ques. – scientist ask – you too
- 1st – try – ask – scientific ques. – diff. things – change way – *puri* – puff up
- Ans. this – simple exp. – try – find – 2 main things – what – change, control – do exp. – what – obs. – changes – any diff.
- This case – think – change – thickness, size – rolled dough
- Also try – diff. flour – *atta*, *maida* – frying *puri* – also change – temp. – hot oil – OR – try – drop – rolled dough – diff. ways
- Make sense – changes – think – what – measure, obs. – some – yes / no ans. – others – no. – measure

- Start – check – *puri* – puffs up – (yes / no) – measure time – puff up (seconds)
- Check – thick layer – dough – thin side – *puri*
- Such exp. – better – change – 1 thing – keep other conditions – same
- Ex. – study effect – boiling hot, hot, not very hot oil – dough – same thickness – dropped – same way
- Also – good idea – keep notes – everything – see, sense – exp.
- After 1 round – think – more ques. – puff better – fresh, stored dough
- Exactly – all scientific exp. – simple to most complicated
- Simple everyday obs. – *puri* puffing – not understood – scientist
- Whether – *puri* puffing – Moon phases – *purnima* – careful obs. – guide you – exploration – investigative world – science